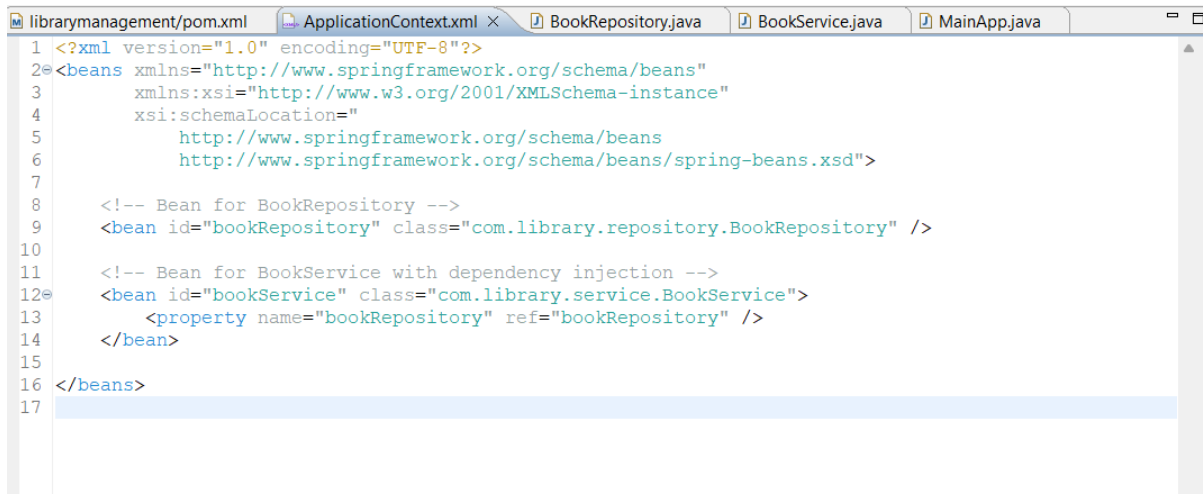
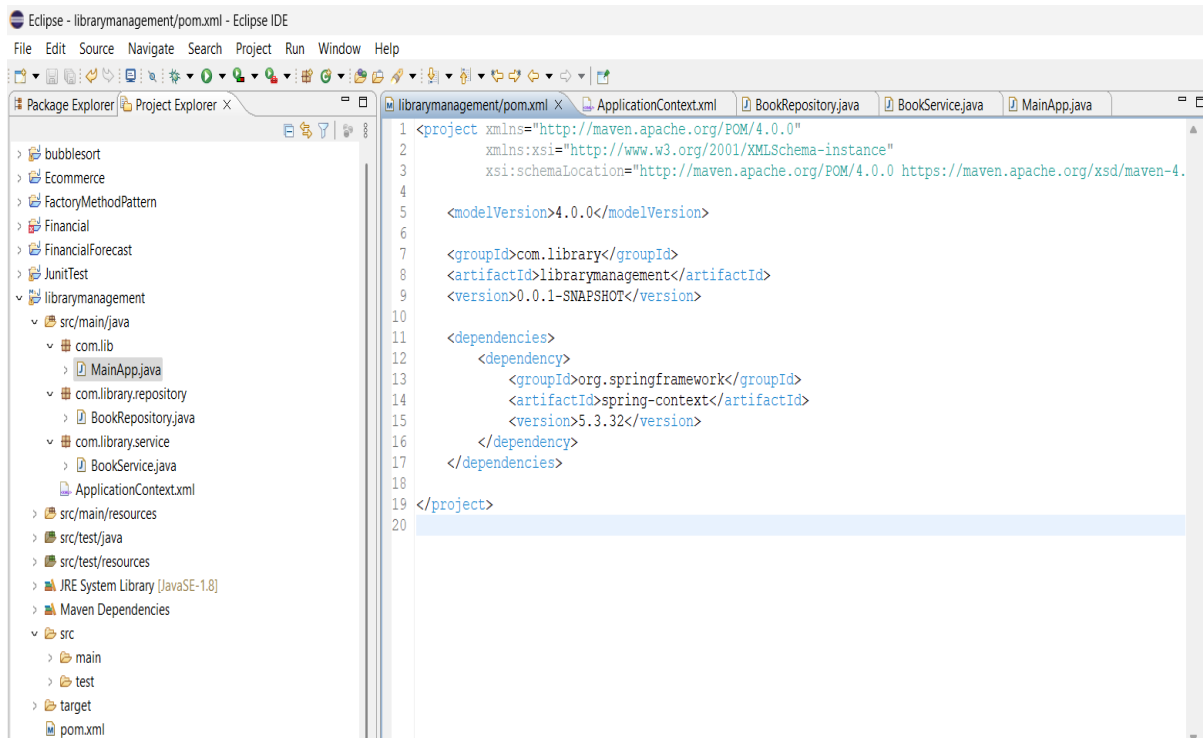


Created By: Shivakant Kurmi

Exercise 1: Configuring a Basic Spring Application



Eclipse - librarymanagement/src/main/java/com/library/repository/BookRepository.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer Project Explorer X

- bubblesort
- Ecommerce
- FactoryMethodPattern
- Financial
- FinancialForecast
- JUnitTest
- librarymanagement
 - src/main/java
 - com.lib
 - MainApp.java
 - com.library.repository
 - BookRepository.java

```
1 package com.library.repository;
2
3 public class BookRepository {
4     public void listBooks() {
5         System.out.println("Listing all books in the library...");
6     }
7 }
8
9
```

librarymanagement/pom.xml ApplicationContext.xml BookRepository.java BookService.java X MainApp.java

```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6     private BookRepository bookRepository;
7
8     // Setter for Spring injection
9     public void setBookRepository(BookRepository bookRepository) {
10         this.bookRepository = bookRepository;
11     }
12
13     public void displayBooks() {
14         bookRepository.listBooks();
15     }
16 }
17
```

```
librarymanagement/pom.xml | ApplicationContext.xml | BookRepository.java | BookService.java | MainApp.java x
1 package com.lib;
2 import com.library.service.BookService;
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 public class MainApp {
7     public static void main(String[] args) {
8         @SuppressWarnings("resource")
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
10
11         BookService service = context.getBean("bookService", BookService.class);
12         service.displayBooks();
13     }
14 }
15
16
17
18
```

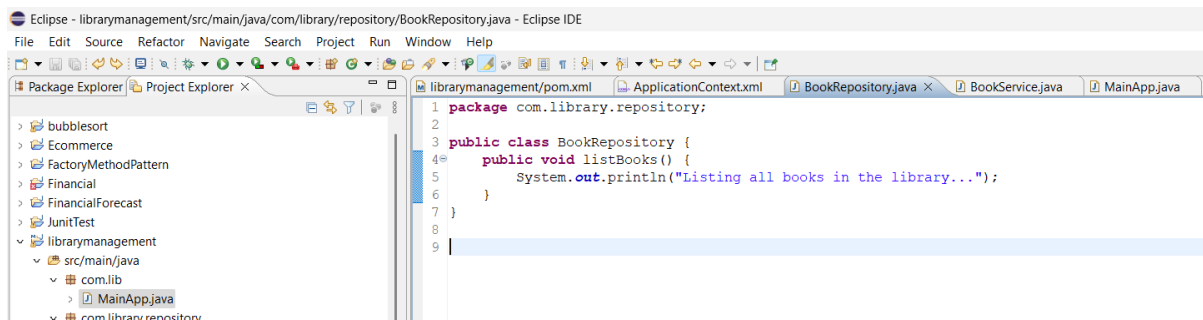
OUTPUT

```
Problems | Javadoc | Declaration | Console x | Coverage
<terminated> MainApp [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Jul 4, 2025, 9:41:28 PM - 9:41:30 PM) [pid: 19704]
Listing all books in the library...

Writable Smart Insert 18:1:540
```

Exercise 2: Implementing Dependency Injection

- ▼ librarymanagement
 - ▼ src/main/java
 - ▼ com.lib
 - ▼ MainApp.java
 - ▼ com.library.repository
 - ▼ BookRepository.java
 - ▼ com.library.service
 - ▼ BookService.java
 - ▼ ApplicationContext.xml
 - > src/main/resources
 - > src/test/java
 - > src/test/resources
 - > JRE System Library [JavaSE-1.8]
 - > Maven Dependencies
 - ▼ src
 - > main
 - > test
 - > target
 - ▼ pom.xml



The screenshot shows the Eclipse IDE interface. The Package Explorer on the left displays the project structure, with 'librarymanagement' expanded to show 'src/main/java/com/library/repository'. The main editor window shows the 'BookRepository.java' file with the following code:

```
1 package com.library.repository;
2
3 public class BookRepository {
4     public void listBooks() {
5         System.out.println("Listing all books in the library...");
6     }
7 }
8
9
```

e/BookService.java - Eclipse IDE

Run Window Help

librarymanagement/pom.xml ApplicationContext.xml BookRepository.java BookService.java MainApp.java

```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6     private BookRepository bookRepository;
7
8     // Setter for Dependency injection
9     public void setBookRepository(BookRepository bookRepository) {
10         this.bookRepository = bookRepository;
11     }
12
13     public void displayBooks() {
14         bookRepository.listBooks();
15     }
16 }
17
```

librarymanagement/pom.xml ApplicationContext.xml BookRepository.java BookService.java MainApp.java

```
1 package com.lib;
2 import com.library.service.BookService;
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 public class MainApp {
7     public static void main(String[] args) {
8         @SuppressWarnings("resource")
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml")
10
11         BookService service = context.getBean("bookService", BookService.class);
12         service.displayBooks();
13     }
14 }
15
16
17
18
```

OUTPUT

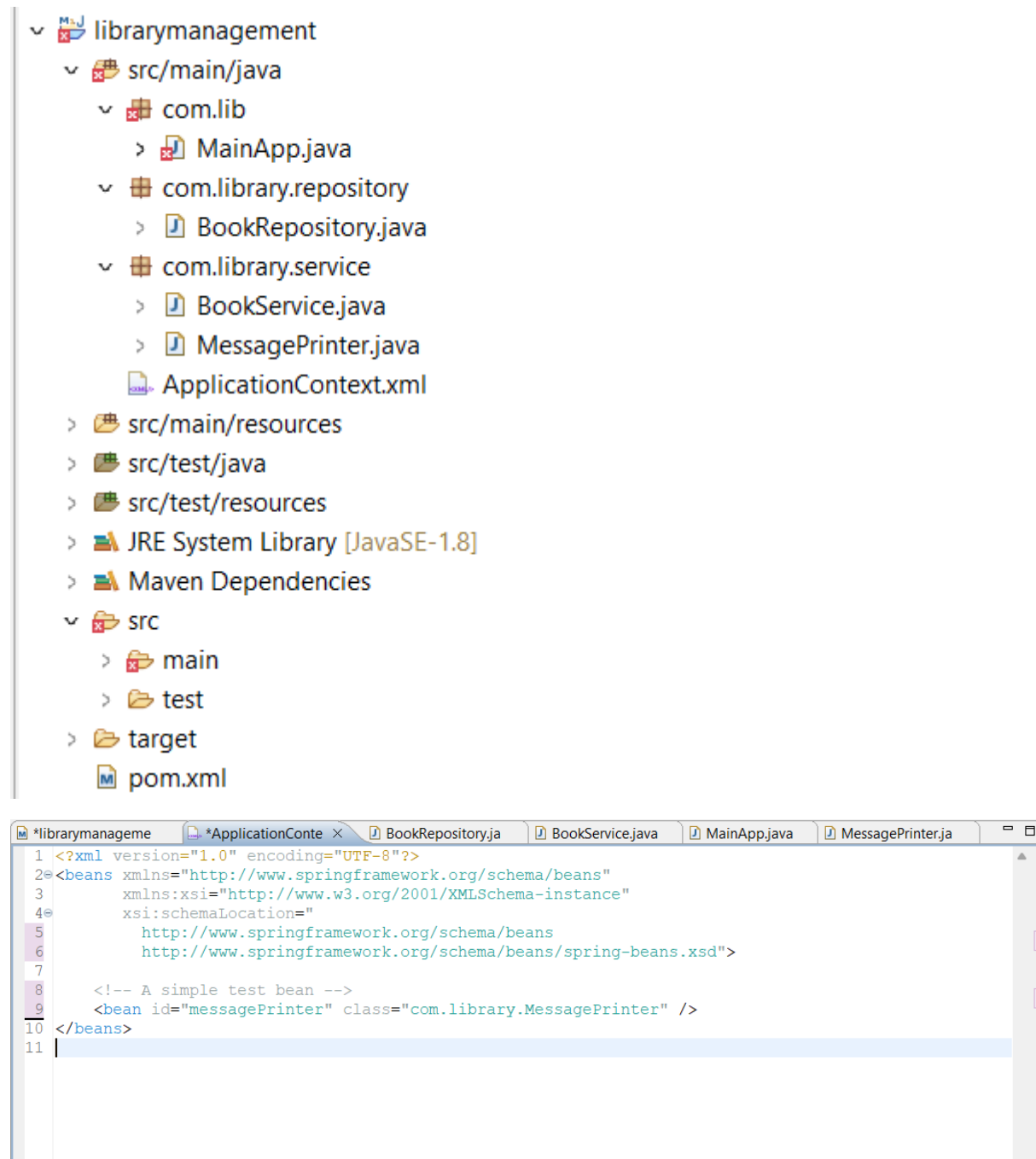
Problems Javadoc Declaration Console Coverage

<terminated> MainApp [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Jul 4, 2025, 10:31:59 PM - 10:32:00 PM) [pid: 8892]

Listing all books in the library...

Writable Smart Insert 18:5:541

Exercise 4: Creating and Configuring a Maven Project



The screenshot displays an IDE interface. The top portion shows a project tree for a Maven project named 'librarymanagement'. The tree structure is as follows:

- librarymanagement
 - src/main/java
 - com.lib
 - MainApp.java
 - com.library.repository
 - BookRepository.java
 - com.library.service
 - BookService.java
 - MessagePrinter.java
 - ApplicationContext.xml
 - src/main/resources
 - src/test/java
 - src/test/resources
 - JRE System Library [JavaSE-1.8]
 - Maven Dependencies
 - src
 - main
 - test
 - target
 - pom.xml

The bottom portion of the screenshot shows the 'ApplicationContext.xml' file being edited. The code in the file is:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <beans xmlns="http://www.springframework.org/schema/beans"
3       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4       xsi:schemaLocation="
5         http://www.springframework.org/schema/beans
6         http://www.springframework.org/schema/beans/spring-beans.xsd">
7
8   <!-- A simple test bean -->
9   <bean id="messagePrinter" class="com.library.MessagePrinter" />
10 </beans>
11
```

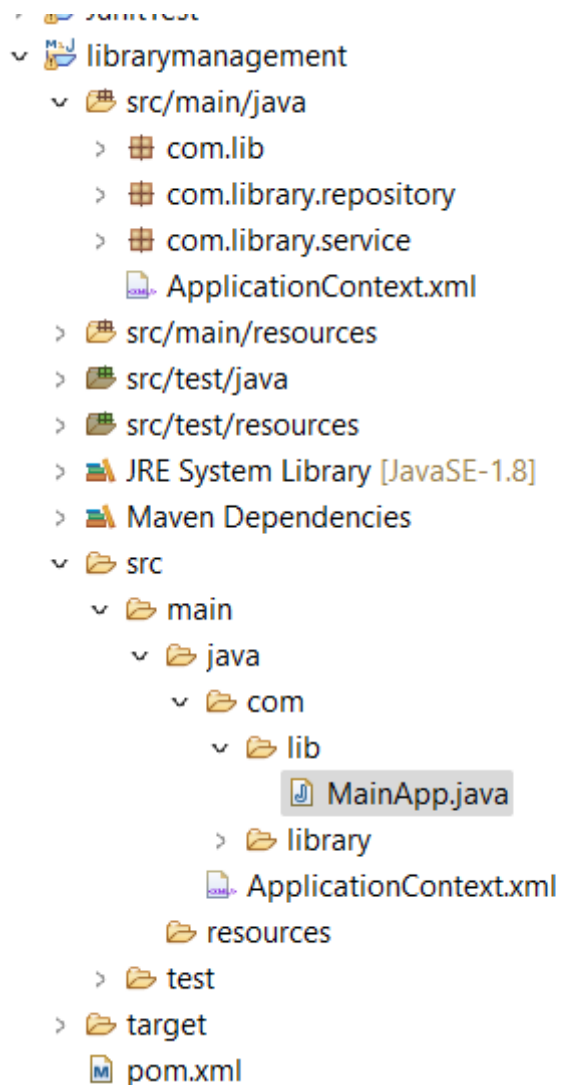
```
*librarymanageme  *ApplicationConte  BookRepository.java  BookService.java  MainApp.java  MessagePrinter.java x
1 package com.library.service;
2 public class MessagePrinter {
3     public void printMessage() {
4         System.out.println("☑ Spring is working! MessagePrinter bean has been created.");
5     }
6 }
7 }
8
```

```
*librarymanageme  *ApplicationConte  BookRepository.java  BookService.java  MainApp.java x  MessagePrinter.java
1 package com.lib;
2 import com.library.service.MessagePrinter;
3
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     public static void main(String[] args) {
9         @SuppressWarnings("resource")
10         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml")
11         // Get the test bean
12         MessagePrinter printer = context.getBean("messagePrinter", MessagePrinter.class);
13         printer.printMessage();
14     }
15 }
16
17
18
19
```

OUTPUT

```
Problems  Javadoc  Declaration  Console x  Coverage
<terminated> Main (3) [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Jul 4, 2025, 10:52:28 PM – 10:52:29 PM) [pid: 6308]
☑ Spring is working! MessagePrinter bean has been created.
```

Exercise 5: Configuring the Spring IoC Container




```
librarymanagement/po  ApplicationContext.x  BookRepository.java  BookService.java  MainApp.java  Main.java
1 <?xml version="1.0" encoding="UTF-8"?>
2 <beans xmlns="http://www.springframework.org/schema/beans"
3     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4     xsi:schemaLocation="
5         http://www.springframework.org/schema/beans
6         http://www.springframework.org/schema/beans/spring-beans.xsd">
7
8     <!-- Define BookRepository bean -->
9     <bean id="bookRepository" class="com.library.repository.BookRepository" />
10
11     <!-- Define BookService bean and inject BookRepository -->
12     <bean id="bookService" class="com.library.service.BookService">
13         <property name="bookRepository" ref="bookRepository" />
14     </bean>
15
16 </beans>
17
```

```
librarymanagement/po  ApplicationContext  BookRepository.java  BookService.java  MainApp.java  Main.java
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6     private BookRepository bookRepository;
7
8     // Setter for Spring DI
9     public void setBookRepository(BookRepository bookRepository) {
10         this.bookRepository = bookRepository;
11     }
12
13     public void displayBooks() {
14         bookRepository.listBooks(); // Call a method to test
15     }
16 }
```

```
librarymanagement/po  ApplicationContext  BookRepository.java  BookService.java  MainApp.java  Main.java
1 package com.library.repository;
2
3 public class BookRepository {
4     public void listBooks() {
5         System.out.println("Listing all books in the library...");
6     }
7 }
8
9
```

```
1 package com.lib;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     @SuppressWarnings("resource")
9     public static void main(String[] args) {
10         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
11
12         // Get bean and call method
13         BookService service = context.getBean("bookService", BookService.class);
14         service.displayBooks();
15     }
16 }
17
18
```

OUTPUT

```
<terminated> MainApp [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Jul 5, 2025, 10:11:38 AM - 10:11:39 AM) [pid: 19120]
Listing all books in the library...
```

Exercise 7: Implementing Constructor and Setter Injection

```
librarymanagement/po  ApplicationContext.x  BookRepository.java  BookService.java  MainApp.java  Main.java
1 <?xml version="1.0" encoding="UTF-8"?>
2 <beans xmlns="http://www.springframework.org/schema/beans"
3     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4     xsi:schemaLocation="
5         http://www.springframework.org/schema/beans
6         http://www.springframework.org/schema/beans/spring-beans.xsd">
7
8     <!-- BookRepository bean -->
9     <bean id="bookRepository" class="com.library.repository.BookRepository" />
10
11     <!-- BookService bean with constructor and setter injection -->
12     <bean id="bookService" class="com.library.service.BookService">
13         <!-- Constructor Injection -->
14         <constructor-arg value="Library Book Service" />
15         <!-- Setter Injection -->
16         <property name="bookRepository" ref="bookRepository" />
17     </bean>
18
19 </beans>
20
```

```
librarymanagement/po  ApplicationContext.x  BookRepository.java  BookService.java  MainApp.java  Main.java
1 package com.library.repository;
2
3 public class BookRepository {
4     public void listBooks() {
5         System.out.println("Listing all books from the bookrepository...");
6     }
7 }
8
9
```

```
librarymanagement/po  ApplicationContext.x  BookRepository.java  BookService.java x  MainApp.java  Main.java
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6     private BookRepository bookRepository;
7     private String serviceName;
8
9     // Constructor for constructor injection
10    public BookService(String serviceName) {
11        this.serviceName = serviceName;
12        System.out.println("Constructor injection: serviceName = " + serviceName);
13    }
14
15    // Setter for setter injection
16    public void setBookRepository(BookRepository bookRepository) {
17        this.bookRepository = bookRepository;
18    }
19
20    public void displayBooks() {
21        System.out.println("Service: " + serviceName);
22        bookRepository.listBooks();
23    }
24 }
```

```
librarymanagement/po  ApplicationContext.x  BookRepository.java  BookService.java  MainApp.java x  Main.java
1 package com.lib;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     @SuppressWarnings("resource")
9     public static void main(String[] args) {
10         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
11
12         BookService service = context.getBean("bookService", BookService.class);
13         service.displayBooks();
14     }
15 }
16
17
18
```

OUTPUT

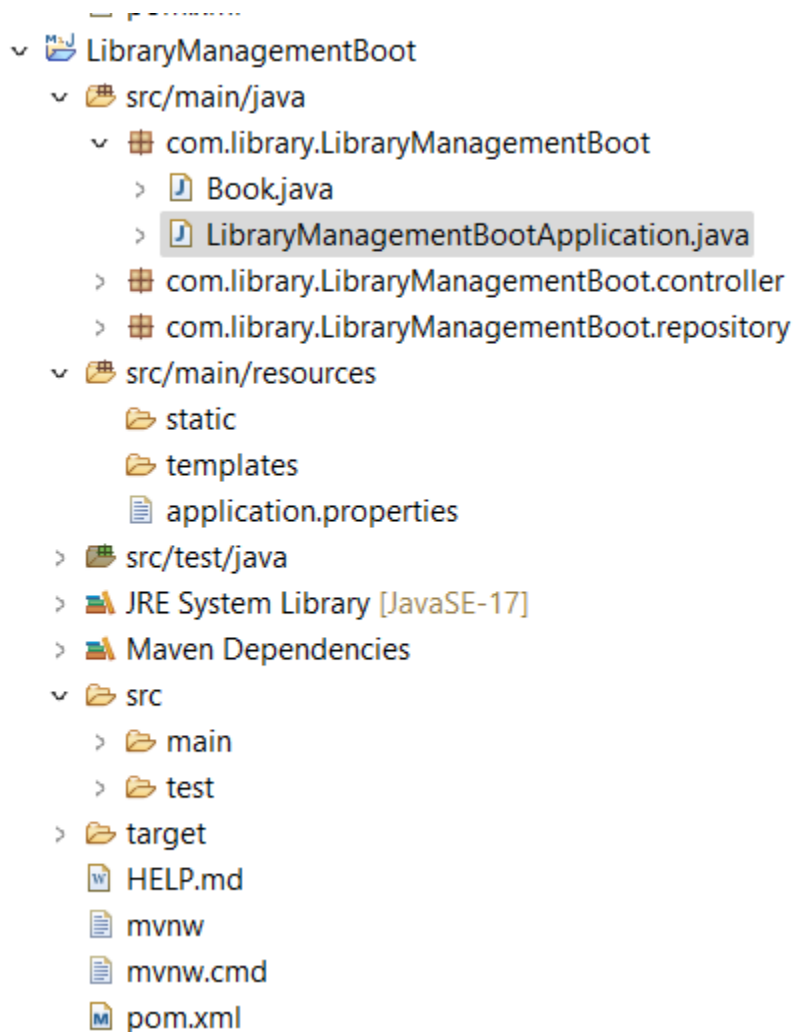
```
Problems  Javadoc  Declaration  Console x  Coverage
<terminated> MainApp [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Jul 5, 2025, 10:18:28 AM - 10:18:32 AM) [pid: 16012]
Constructor injection: serviceName = Library Book Service
Service: Library Book Service
Listing all books from the bookrepository...
```

Exercise 9: Creating a Spring Boot Application

The screenshot shows the Spring Initializr web application interface. The browser tabs include 'Spring Initializr', 'localhost:8082/books', and 'H2 Console'. The address bar shows 'https://start.spring.io'. The interface is divided into several sections:

- Project:** Includes radio buttons for 'Gradle - Groovy', 'Gradle - Kotlin', and 'Maven' (selected).
- Language:** Includes radio buttons for 'Java' (selected), 'Kotlin', and 'Groovy'.
- Spring Boot:** Includes radio buttons for versions: '4.0.0 (SNAPSHOT)', '3.5.4 (SNAPSHOT)', '3.5.3' (selected), '3.4.8 (SNAPSHOT)', '3.4.7', and '3.3.13'.
- Project Metadata:** Includes text input fields for 'Group' (com.library), 'Artifact' (LibraryManagementBoot), 'Name' (LibraryManagementBoot), 'Description' (Demo project for Spring Boot), and 'Package name' (com.library.LibraryManagementBoot).
- Packaging:** Includes radio buttons for 'Jar' (selected) and 'War'.
- Java:** Includes radio buttons for versions: '24', '21', and '17' (selected).
- Dependencies:** Includes a section for adding dependencies with a button 'ADD DEPENDENCIES... CTRL + B'. Below this, there are three dependency cards: 'Spring Web' (WEB), 'Spring Data JPA' (SQL), and 'H2 Database' (SQL).

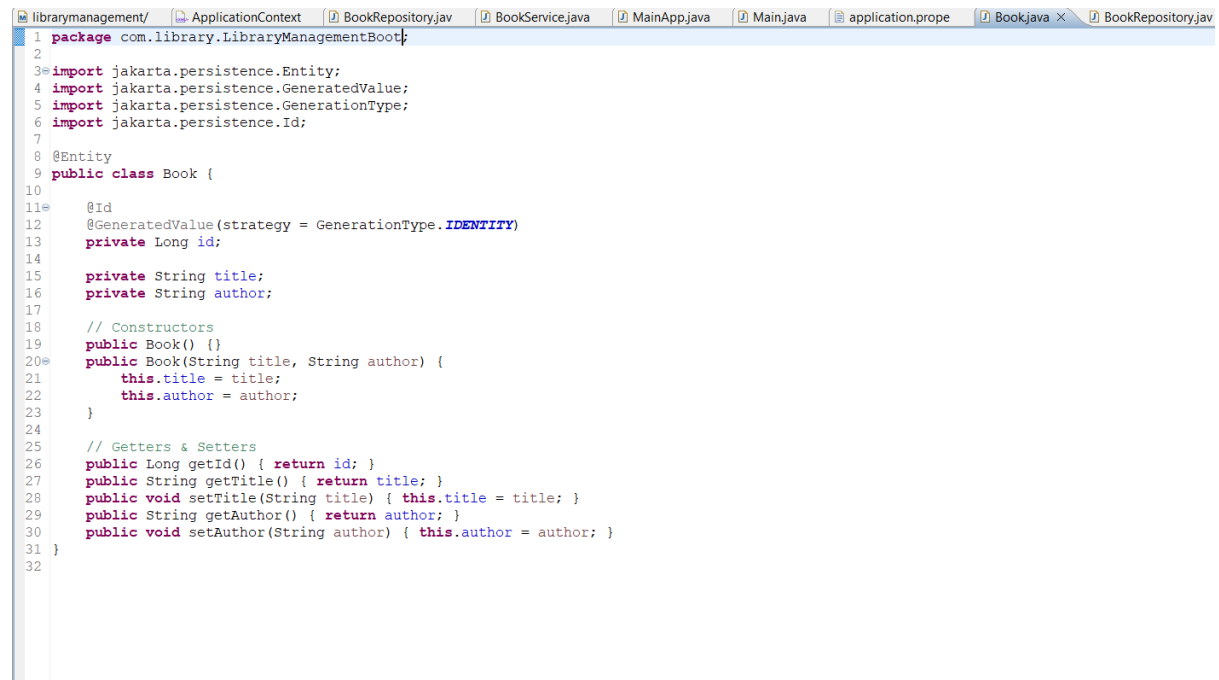
At the bottom, there are three buttons: 'GENERATE CTRL + G', 'EXPLORE CTRL + SPACE', and an ellipsis button '...'.



Application.process.context

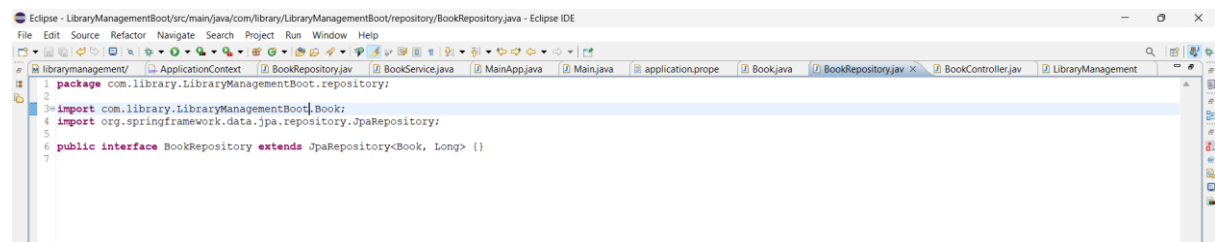
```
1 spring.application.name=LibraryManagementBoot
2 # H2 in-memory DB config
3 spring.datasource.url=jdbc:h2:mem:librarydb
4 spring.datasource.driverClassName=org.h2.Driver
5 spring.datasource.username=sa
6 spring.datasource.password=
7 server.port=8082
8
9 # JPA config
10 spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
11 spring.jpa.hibernate.ddl-auto=update
12
13 # H2 console
14 spring.h2.console.enabled=true
15 spring.h2.console.path=/h2-console
16
```

Book.java

A screenshot of the Eclipse IDE showing the Book.java file. The package is com.library.LibraryManagementBoot. The class Book is an entity with fields id, title, and author. It has two constructors and several getters and setters.

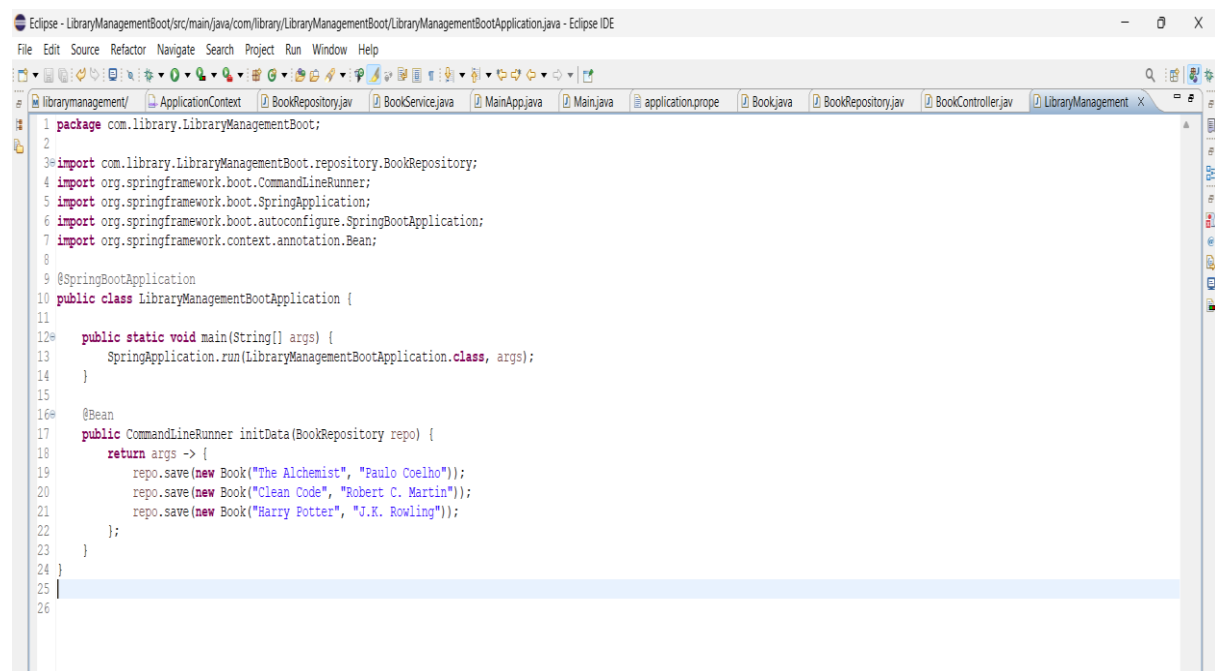
```
1 package com.library.LibraryManagementBoot;
2
3 import jakarta.persistence.Entity;
4 import jakarta.persistence.GeneratedValue;
5 import jakarta.persistence.GenerationType;
6 import jakarta.persistence.Id;
7
8 @Entity
9 public class Book {
10
11     @Id
12     @GeneratedValue(strategy = GenerationType.IDENTITY)
13     private Long id;
14
15     private String title;
16     private String author;
17
18     // Constructors
19     public Book() {}
20     public Book(String title, String author) {
21         this.title = title;
22         this.author = author;
23     }
24
25     // Getters & Setters
26     public Long getId() { return id; }
27     public String getTitle() { return title; }
28     public void setTitle(String title) { this.title = title; }
29     public String getAuthor() { return author; }
30     public void setAuthor(String author) { this.author = author; }
31 }
32
```

Bookrepository.java

A screenshot of the Eclipse IDE showing the BookRepository.java file. The package is com.library.LibraryManagementBoot.repository. The interface BookRepository extends JpaRepository<Book, Long>.

```
1 package com.library.LibraryManagementBoot.repository;
2
3 import com.library.LibraryManagementBoot.Book;
4 import org.springframework.data.jpa.repository.JpaRepository;
5
6 public interface BookRepository extends JpaRepository<Book, Long> {}
7
```

LibraryManagement

A screenshot of the Eclipse IDE showing the LibraryManagementBootApplication.java file. The package is com.library.LibraryManagementBoot. The class LibraryManagementBootApplication is a Spring Boot application with a main method and an @Bean method initData.

```
1 package com.library.LibraryManagementBoot;
2
3 import com.library.LibraryManagementBoot.repository.BookRepository;
4 import org.springframework.boot.CommandLineRunner;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7 import org.springframework.context.annotation.Bean;
8
9 @SpringBootApplication
10 public class LibraryManagementBootApplication {
11
12     public static void main(String[] args) {
13         SpringApplication.run(LibraryManagementBootApplication.class, args);
14     }
15
16     @Bean
17     public CommandLineRunner initData(BookRepository repo) {
18         return args -> {
19             repo.save(new Book("The Alchemist", "Paulo Coelho"));
20             repo.save(new Book("Clean Code", "Robert C. Martin"));
21             repo.save(new Book("Harry Potter", "J.K. Rowling"));
22         };
23     }
24 }
25
26
```

OUTPUT

```
Problems Javadoc Declaration Console Coverage
LibraryManagementBootApplication [Java Application] [pid: 19344]

: Spring Boot :: (v3.5.3)

25-07-05T11:07:05.133+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:05.137+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:06.180+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:06.288+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.030+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.051+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.126+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.128+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.472+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.565+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.615+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.945+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:07.977+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:08.188+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:08.191+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:08.231+05:30 WARN 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:08.275+05:30 INFO 19344 --- [LibraryManagementBoot] [
Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 2.3.232
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
25-07-05T11:07:09.057+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:09.119+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:09.461+05:30 WARN 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:09.914+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:10.084+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:07:10.101+05:30 INFO 19344 --- [LibraryManagementBoot] [
25-07-05T11:09:14.390+05:30 INFO 19344 --- [LibraryManagementBoot] [nio-8082-exec-1]

main c.l.l.LibraryManagementBootApplication : Starting LibraryManagementBootApplication using Java 21
main c.l.l.LibraryManagementBootApplication : No active profile set, falling back to 1 default profile
main .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode
main .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 93 ms. Found 0
main o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8082 (http)
main o.apache.catalina.core.StandardService : Starting service [Tomcat]
main o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
main o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
main w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1.1
main o.hibernate.jpa.internal.util.LogHelper : HHH0000204: Processing PersistenceUnitInfo [name: default]
main org.hibernate.Version : HHH0000412: Hibernate ORM core version 6.6.18.Final
main o.h.c.internal.RegionFactoryInitiator : HHH0000026: Second-level cache disabled
main o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
main com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
main com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:
main com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
main org.hibernate.orm.deprecation : HHH90000025: H2Dialect does not need to be specified explicitly
main org.hibernate.orm.connections.pooling : HHH10001005: Database info:

o.h.e.t.j.p.i.JtaPlatformInitiator : HHH0000489: No JTA platform available (set 'hibernate.transaction
j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit
JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore
o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available
o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8082 (http) with context path '/'
c.l.l.LibraryManagementBootApplication : Started LibraryManagementBootApplication in 5.518 seconds
o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
```

Localhost:8082/books

```
Spring Initializr localhost:8082/books

localhost:8082/books

Pretty-print

[{"id":1,"title":"The Alchemist","author":"Paulo Coelho"}, {"id":2,"title":"Clean Code","author":"Robert C. Martin"}, {"id":3,"title":"Harry Potter","author":"J.K. Rowling"}]
```

Localhost:8082/h2-console

Spring Initializr localhost:8082/books H2 Console

localhost:8082/h2-console/login.do?sessionId=40094fd7c0bb5bb7e851604d4e364c07

English Preferences Tools Help

Login

Saved Settings: Generic H2 (Embedded)

Setting Name: Generic H2 (Embedded) Save Remove

Driver Class: org.h2.Driver

JDBC URL: jdbc:h2:~test

User Name: sa

Password:

Connect Test Connection

